

Realtime Chat Application

Comprehensive System Documentation, Architecture & New Feature Walkthrough

MERN Stack

Socket.io (WebSockets)

Tailwind CSS + Vite

JWT Dual-Token Security

Photo & Video Sharing

Friend Request System

Multi-User Groups

1. Project Overview & Architecture

Ye application ek **Full-Stack Production-Ready Realtime Chat System** hai. Decoupled frontend aur backend ke saath, ye **HTTP REST APIs** aur **Bidirectional WebSockets (Socket.io)** ke through aapas me interact karta hai.

2. Newly Implemented Features (Highlights)

NEW FEATURE 1. Photo & Video Sharing

- Users text messages ke saath-saath **high-resolution Photos aur Videos** share kar sakte hain.
- **Built-in Video Player:** Chat bubbles me video player embed hota hai jisme play/pause, volume aur fullscreen controls hote hain.
- **Dual Storage Engine:** Cloudinary configured hone par CDN use karta hai; agar Cloudinary keys na ho toh automatically local disk storage (/uploads) use karta hai. Photos aur Videos locally 100% chalte hain!

NEW FEATURE 2. Friend Request System (Protected 1-on-1 DMs)

- Koi bhi user kisi anjaan user ko direct message (DM) nahi kar sakta jab tak dono **Friends** na ban jayein.
- **User Search:** Username se users ko khojkar Friend Request bhejne ki suvidha.
- **Incoming & Outgoing Requests:** Requests tab me incoming requests ko *Accept* ya *Decline* kar sakte hain, aur outgoing requests ko *Cancel* kar sakte hain.
- **Real-time Notification:** Friend request aane par recipient ko instant socket alert aur notification badge milta hai.

NEW FEATURE 3. Multi-User Group Creation

- Users ek se zyada dosto ko select karke **Custom Groups** bana sakte hain.
- Group Name, Description aur Multi-Select friends list ke through group create hota hai.
- Group members ko instantly notification milti hai aur group sidebar me 👥 icon ke saath live show hota hai.

NEW FEATURE 4. Handcrafted Light Mode & Dark Mode Switcher

- Seamless 1-click ☀️ / 🌙 theme switch jo Sidebar aur Auth dono jagah available hai.
- **Local Persistence:** User ka chosen theme localStorage me save rehta hai.
- **Handcrafted Aesthetics:** Crisp slate-navy typography, soft card borders, glassmorphic shadows, aur tailored light tokens taaki app bilkul authentic aur bespoke product lage.

3. Technology Breakdown: Kya, Kyu, Kaise aur Kaha

Backend Architecture (server/)

MODULE / FILE	KYA USE KIYA?	KYU USE KIYA?	KAISE KAAM KARTA HAI?
models/User.model.js	Mongoose Schema + friends array	User identity, password hashing (bcrypt), aur friends relationship store karne ke liye.	Pre-save hook me 12 salt rounds se hashing. friends array me accepted friend IDs store hoti hain.
models/FriendRequest.model.js	Compound Index Schema	Pending, Accepted, aur Rejected requests manage karne ke liye.	Unique compound index { sender: 1, recipient: 1 } duplicate requests rokta hai.
models/Room.model.js	isGroup, members, isPrivate	Channels, 1-on-1 DMs, aur Multi-user Groups ko ek hi collection me support karne ke liye.	Channels: public. DMs: private (2 members). Groups: private (multiple selected members).
models/Message.model.js	type: 'image' 'video' 'file'	Multimedia chat messages aur replies store karne ke liye.	fileUrl, fileType, fileName store karta hai. Compound index fast loading deta hai.
controllers/friend.controller.js	Friend Request Logic	Friend requests send, accept, reject, list aur user discovery ke liye.	Accept hone par dono users ko ek doosre ke friends array me add karta hai aur socket event emit karta hai.
middleware/upload.middleware.js	Multer + Dual Storage	Photos aur videos ko 50MB tak upload karne ke liye.	Cloudinary keys ho toh Cloudinary, warna local public/uploads me save karke static serve karta hai.

Frontend Architecture (client/)

COMPONENT / FILE	KYA KAAM KARTA HAI?	KYU ZARURI HAI?
components/chat/FriendsModal.jsx	Friends list, Incoming/Outgoing requests, User search & Add Friend tab.	Users ko dosto se connect karne aur unke saath private chat start karne ke liye.
components/chat/CreateGroupModal.jsx	Group name, description, multi-select checkboxes for friends.	Ek sath multiple friends ka group chat room banane ke liye.
components/chat/MessageBubble.jsx	Photos, Videos (HTML5 Video Player), Text, Emojis, Replies	Video aur photo playback bina page reload kiye chat ke andar

COMPONENT / FILE	KYA KAAM KARTA HAI?	KYU ZARURI HAI?
	display karta hai.	direct render karta hai.
<code>components/chat/MessageInput.jsx</code>	File picker, attachment preview thumbnail/badge, caption support, debounced typing.	Send karne se pehle attachment ka preview dikhata hai aur send hone par broadcast karta hai.
<code>components/layout/Sidebar.jsx</code>	Channels, Groups, Direct Messages, aur Friends button with live request badge.	Sleek dark-mode navigation structure jisme alag-alag conversations organized rehti hain.

4. Running & Testing Instructions

1. **Backend:** `cd server && npm run dev` (Port 5000)

2. **Frontend:** `cd client && npm run dev` (Port 5173)

3. Testing Steps:

- Browser me `http://localhost:5173` par do alag-alag accounts register karein (ek normal tab me, ek incognito me).
- **Friends & Requests** button par click karke dusre user ko Friend Request bhejin aur dusre tab me *Accept* karein.
- Accept hone ke baad **Direct Message** start karein, photos aur videos upload karke live preview test karein!
- **+ Groups** par click karke group ka name dalein aur apne dost ko add karke Multi-User Group create karein!